Accumulation of computer rounding errors (1.45)

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Accumulation of your computer rounding errors. Start with $x_0 = 1$, $y_0 = 0$, and, with your computer, generate the sequences

$$x_{n+1} = \frac{5x_n - 12y_n}{13}, \qquad y_{n+1} = \frac{12x_n + 5y_n}{13}.$$

Find $x_n^2 + y_n^2$ for $n = 10^2, 10^3, 10^4, 10^5, 10^6$, and 10^7.